Probability and Statistics (XB 0115)

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Why should you learn probability and statistics?

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https://www.mathmatters.ai/

Strong Math Foundations are Important for AI

Artificial Intelligence is poised to transform society as we know it. To be prepared for this future, it is imperative we educate our future workforce with the knowledge to build and deploy AI technology. **Core mathematical concepts from algebra, calculus, and probability lie at the heart of modern AI innovation.** *As such, engaging in the development of these technologies requires that students start with strong mathematical foundations.*

Why should you learn probability and statistics?

While today's advances might suggest classic mathematical topics like calculus or algebra are outdated, nothing could be further from the truth. In reality, modern AI systems are rooted in mathematics, making a strong command over math necessary for careers in this field.

Studying these core topics in high school is the best preparation for later specialization in machine learning, data science, or any STEM field, and generally **we would much rather hire students who have mastered fundamentals than those who have a shallow familiarity with the latest tools or software.**

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What is Probability Theory?

- Randomness and uncertainty exist in our daily lives and in science, engineering and technology.
- We require a mathematical framework that allows us to analyze random phenomena.
- **Probability theory** provides us such a framework.
- But what do we mean by **random phenomena** and **probability**?

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Randomness

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A **random experiment** is an experiment[a] with an uncertain outcome.

[a] In a broad sense: experiment, process, event, action, phenomenon.

Examples:

- Throwing a die or coin
- Diagnosing an illness
- The outside temperature in one week
- Next U.S. president
- Occurrence of wild fires
- ...

More on Randomness

- We can think about randomness as a way to express *what we do not know* (i.e. our lack of knowledge)
- Let's go back to the coin experiment. What if we knew more about the force with which the coin was flipped, the initial orientation, etc.
- With all the information provided, we might be able to predict the outcome of the coin flip.
- When our knowledge about the outcome is limited, we say that it is *random*.
- So randomness is not an inherent property of nature, but reflects our lack of knowledge.[1]

[1] Except for quantum mechanical processes.

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Interpretations of Probability

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- If we flip a coin many times, without prior information about the flip, we say the probability of heads is 50% or $\frac{1}{2}$. What does this mean?
- **Probability as relative frequency**: If we flip the coin a large number of times, it will come to heads about $\frac{1}{2}$ the time.
- But what about when we say something like "It is likely that the conflict will escalate."?
- **Probability as subjective belief**: Probability is the quantification of our belief that something will happen.

Interpretations of Probability

- The two interpretations often coincide since personal beliefs are based on the assessment of relative frequency of events.
- The beauty of probability theory is that it is applicable in both cases.
- It provides a solid framework to study random phenomena and starts with the **axioms of probability**.

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Set Theory

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A **set** is a collection of things called **elements**.

- A set is denoted in capital letters and defined by simply listing its elements in curly brackets. Example: $A = \{b, c\}$.
- Ordering does not matter. Thus $\{1, 2, 3, 4\}$ and $\{3, 2, 1, 4\}$ are the same sets.
- A set can also be defined by $B = \{x : x \text{ satisfies some property}\}$. [2] Example: $B = \{x \in \mathbb{N} : x \text{ even}\}$.
- $b \in A$ means that b belongs to A ($\in$ means "is an element of").
- $d \notin$ means that d does not belong to A ($\notin$ means "is not an element of").

[2] Often a vertical bar | is used instead of a colon :.

Important Sets

- The set of natural numbers, $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$
- The set of integers, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
- The set of rational numbers $\mathbb{Q}$.
- The set of real numbers $\mathbb{R}$ and the set of complex numbers $\mathbb{C}$.
- Closed intervals on the real line such as

$$[2, 3] = \{x \in \mathbb{R} : 2 \leq x \leq 3\}.$$

- Open intervals on the real line such as

$$(1, 2) = \{x \in \mathbb{R} : 1 < x < 2\}.$$

- Intervals that are neither open nor closed such as

$$(1, 3] = \{x \in \mathbb{R} : 1 < x \leq 3\}.$$

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More on Sets

- Set A is a **subset** of set B if every member of A is also a member of B. We write $A \subset B$, where $\subset$ mean "is subset of".
- Equivalently B is the **superset** of A, $B \supset A$.
- Two sets are **equal** $A = B$, if they contain the same elements, that is $A \subset B$ and $B \subset A$.
- The **universal set** S or Ω is the set of all things that we could possibly consider in the context we are studying.
- The universal set in probability is also called the **sample space**.
- The set with no elements is called the **empty** or **null set** $\phi = \emptyset$.

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Venn Diagrams

- Venn Diagrams are very useful in visualizing relations between sets.
- In Venn Diagrams, a set is depicted by a closed region.

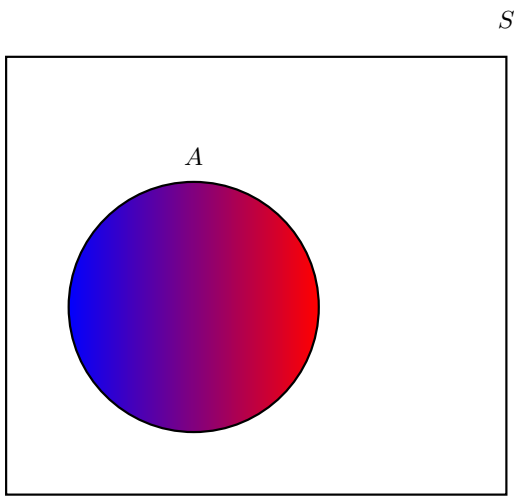

Figure: Venn Diagram

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Venn Diagrams

- The figure below shows two sets, A and B, where $B \subset A$.
- Both A and B are subsets of the universal set S.

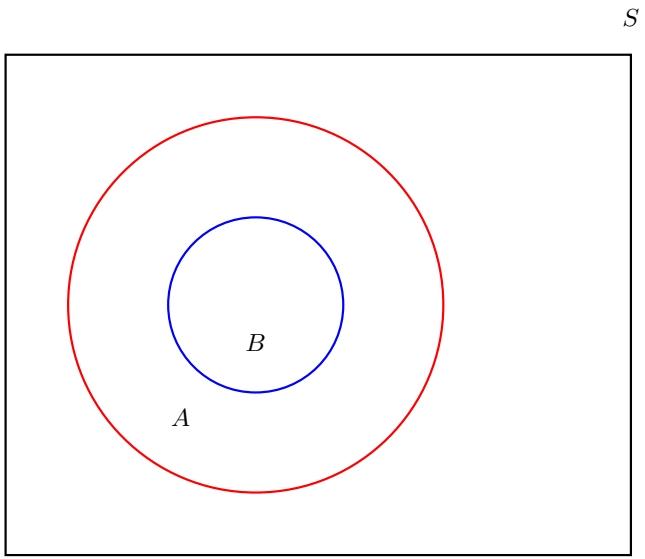

Figure: Venn Diagram for two sets A and B, where $B \subset A$.

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Set Operations: Union

- The union of two sets A and B is a set $A \cup B$ containing all elements that are in A or[3] in B.
- Example: $\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$.
- In general the union of n sets $A_1, A_2, ..., A_n$ is written as $\bigcup_{i=1}^{n} A_i$.

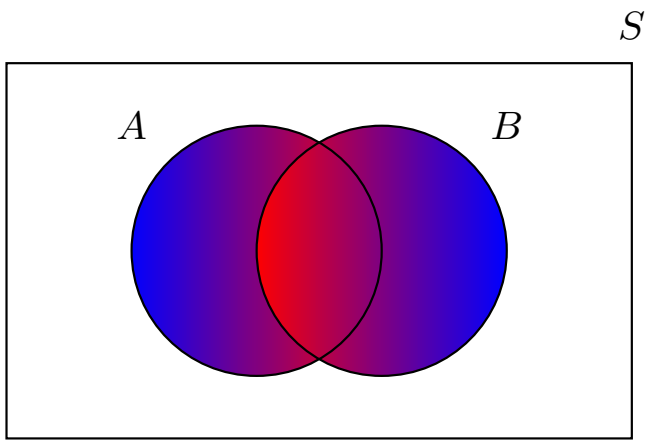

Figure: The shaded area shows the set $A \cup B$.

[3] We use the inclusive or (not the exclusive or).

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Set Operations: Intersection

- The intersection of two sets A and B is a set $A \cap B$ containing all elements that are in A and in B.
- Example: $\{1, 2\} \cap \{2, 3\} = \{2\}$.
- In general, the intersection of n sets $\bigcap_{i=1}^{n} A_i$ is the set consisting of elements that are in all n sets.

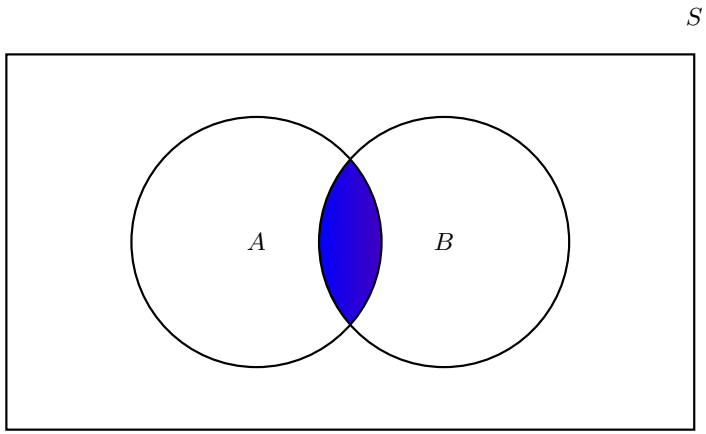

Figure: The shaded area shows the set $A \cap B$.

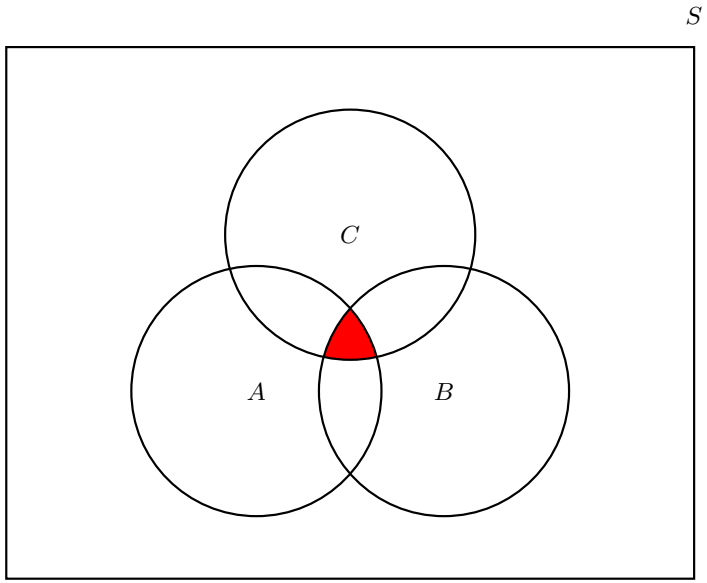

Figure: The shaded area shows the set $A \cap B \cap C$.

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Set Operations: Complement

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- The complement of a set A is the set A^c (also denoted by A' or $\bar{A}$) of all elements that are in the universal set S but not in A.

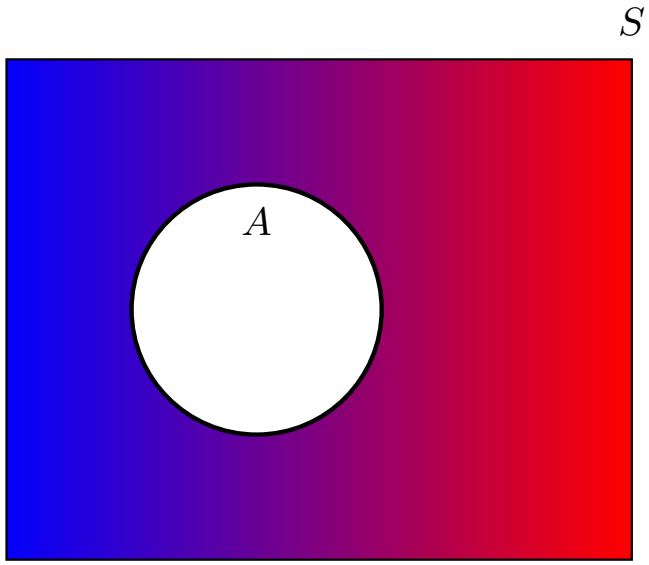

Figure: The shaded area shows the set A^c.

Set Operations: Difference (Subtraction)

- The set $A - B$ (also denoted by A/B) consists of all elements that are in A but not in B.
- Example: $A = \{1, 2, 3\}$ and $B = \{3, 5\}$, then $A - B = \{1, 2\}$

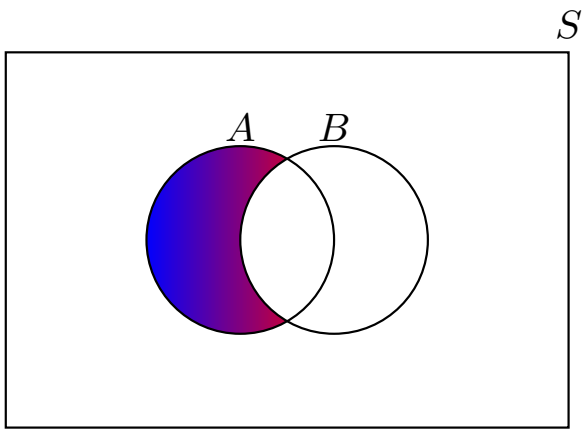

Figure: The shaded area shows the set $A - B$.

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Exercise

Let A, B, C be sets. For each of the following sets, draw a Venn diagram and shade the area representing the given set.

- $A \cup B \cup C$
- $A \cap B \cap C$
- $A \cup (B \cap C)$
- $A - (B \cap C)$
- $A \cup (B \cap C)^c$

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Mutually Exclusive or Disjoint Sets

- Sets A and B are **mutually exclusive** or **disjoint** if they do not have any shared elements.
- This is equivalent to $A \cap B = \emptyset$.
- In general, multiple sets are called disjoint if they are pairwise disjoint.

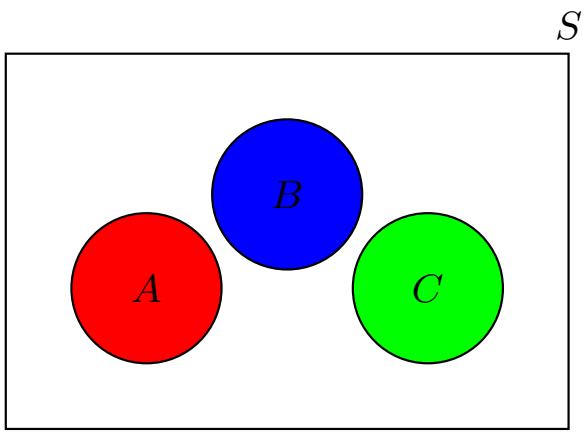

Figure: Sets A, B, and C are disjoint.

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Partition of Sets

- A collection of non-empty sets $A_1, A_2, \ldots$ is a **partition** of S if they are disjoint and their union is S.

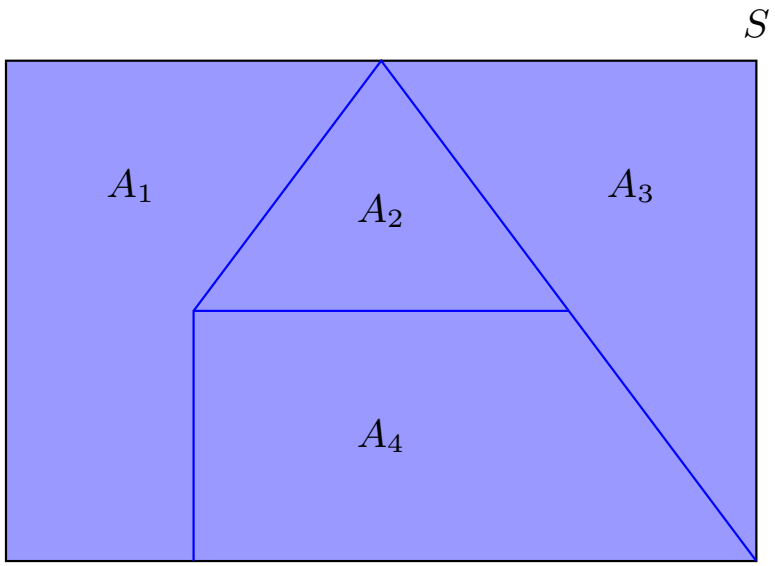

Figure: The collection of sets A_1, A_2, A_3 and A_4 is a partition of S.

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Important Theorems

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De Morgan's laws. Any collection $A_1, A_2, \ldots, A_n$ of sets satisfies

$$(A_1 \cup A_2 \cup \ldots \cup A_n)^c = A_1^c \cap A_2^c \cap \ldots \cap A_n^c$$

$$(A_1 \cap A_2 \cap \ldots \cap A_n)^c = A_1^c \cup A_2^c \cup \ldots \cup A_n^c$$

Distributive laws. Any sets A, B, C satisfy[a]

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

[a] Compare to $a \times (b + c) = a \times b + a \times c$, i.e. multiplication is distributive over addition.

Exercise

Let $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2\}$, $B = \{2, 4, 5\}$, and $C = \{1, 5, 6\}$.

Find the following sets:

- $A \cup B$
- $A \cap B$
- A^c
- B^c
- Check de Morgan's law by finding $(A \cup B)^c$ and $A^c \cap B^c$.
- Check the distributive law by finding $A \cap (B \cup C)$ and $(A \cap B) \cup (A \cap C)$.

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The Cartesian Product

The **Cartesian Product** of two sets A and B is the set

$$A \times B = \{(x, y) : x \in A \text{ and } y \in B\}.$$

- $A \times B$ is the set of *ordered pairs*[4] from A and B.
- For example, if $A = \{1, 2, 3\}$ and $B = \{H, T\}$, then

$$A \times B = \{(1, H), (1, T), (2, H), (2, T), (3, H), (3, T)\}.$$

- Note that the pairs are ordered, thus $(1, H) \neq (H, 1)$ and $A \times B \neq B \times A$.
- Another example: $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is set of all points in the 2-dimensional plane.

[4] So $(x, y) \neq (y, x)$ in general.

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The Multiplication Principle

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- **Multiplication principle**: If A has M elements and B has N elements, then $A \times B$ has $M \times N$ elements. More formally:

$$|A \times B| = |A| \times |B| = M \times N$$

.

- More generally: For sets $A_1, A_2, \ldots A_n$ with $|A_i| = M_i$ for $i = 1, \ldots, n$:

$$|A_1 \times A_2 \times \ldots \times A_n| = M_1 \times M_2 \times \ldots \times M_n$$

.

- Example: When rolling three dice, the total number of possible outcomes is $6 \times 6 \times 6 = 216$.

Cardinality of Finite sets

- The **cardinality** $|A|$ of a set A is basically its size.
- If A only has a finite number of elements, its cardinality is simply the number of elements in A.
- For example, if $A = \{2, 4, 6, 8, 10\}$, then $|A| = 5$.
- We will discuss cardinality of infinite sets later.

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The Inclusion-Exclusion Principle

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The inclusion-exclusion principle. For all finite sets A, C, B:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

More generally: For all finite sets $A_1, A_2, \ldots, A_n$:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Example

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Example

At a party:

- there are 10 people with white shirts and 8 people with red shirts
- 4 people have black shoes and white shirts
- 3 people have black shoes and red shirts
- the total number of people with white or red shirts or black shoes is 21

How many people have black shoes?

Cardinality of Infinite Sets

- There are two kinds of infinite sets: countable sets and uncountable sets.
- The difference between the two is that you can list elements in a countable set, so $A = \{a_1, a_2, \ldots\}$, but you cannot list elements in an uncountable set.
- The set $\mathbb{R}$ is uncountable and much "larger" than countably infinite sets such as $\mathbb{N}$ and $\mathbb{Z}$.

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Countable versus Uncountable Sets

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- A more rigorous definition of a countable set A is: It is either **finite** or **countably infinite**, i.e. it can be put in one-to-one correspondence with $\mathbb{N}$.
- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and any of their subsets are countable and hence "equally large".
- Any set containing an interval on the real line such as $[a, b], (a, b], [a, b)$ and (a, b), where $a < b$ is uncountable.

Useful Theorems on Countability

- Any subset of a countable set is countable. Any superset of an uncountable set is uncountable.
- If $A_1, A_2, \cdots$ is a list of countable sets, then the set $\bigcup_i A_i = A_1 \cup A_2 \cup A_3 \cdots$ is also countable.
- If A and B are countable, then $A \times B$ is also countable.

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Functions

- A **function** maps elements from the **domain** set to elements in another set called the **co-domain**.
- Each input in the domain is mapped to exactly one output in the co-domain.
- It is denoted as $f : A \to B$.
- The **range** of a function is the set of all possible values of $f(x)$ and is a subset of the co-domain.

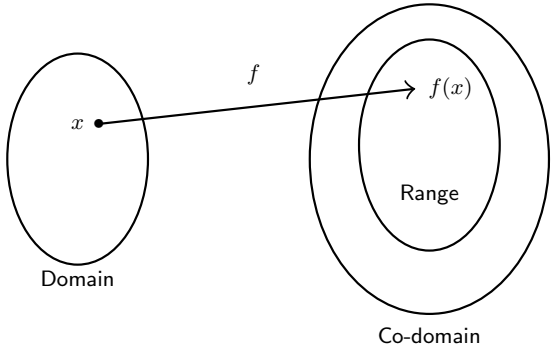

Figure: Function $f : A \to B$, the range is always a subset of the co-domain.

Example

What are the domain, codomain, and range of (i) $f(x) = x^2$ and (ii) $g(x) = \sin x$?

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Exercises

Section 1.5: Exercises 1, 2, 3, 4, 5, 6, 7, 12.

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Random Experiments

A **random experiment** is a process with an uncertain outcome.

An **outcome** is a result of a random experiment.

The set of all possible outcomes is called the **sample space** (S) and in this context is the universal set.

When we repeat a random experiment several times, we call each one a **trial**.

Examples

Throwing a die or a coin, measure a physical quantity (i.e. temperature or pressure), surveys and questionnaires, testing a new medicine, taking an exam,...

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Axioms of Probability

To any **event** $A \subset S$ we want to assign a number $0 \leq P(A) \leq 1$ that measures how likely it is that A occurs:

$P(A)$ close to 0 means "A is unlikely to occur" and $P(A)$ close to 1 means "A is likely to occur".

A function P is a **probability measure** if:

Axiom 1: For any event $A, P(A) \geq 0$

Axiom 2: Probability of the sample space S is $P(S) = 1$

Axiom 3: If $A_1, A_2, A_3 \ldots$ are disjoint events, then

$$P(A_1 \cup A_2 \cup A_3 \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots$$

$P(A)$ is called the **probability of** A.

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Some elementairy properties of probabilities

- $P(A^c) = 1 - P(A)$.
- $P(\phi) = 0$.
- For any event $P(A) \leq 1$.
- If $A \subset B$ then $P(A) \leq P(B)$.
- $P(A/B) = P(A) - P(A \cap B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

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Discrete Probability Models

A **probability model** is the specification of a sample space S and a probability measure P. A probability model is **discrete** if S is countable (otherwise it is **continuous**).

- If S is discrete, we can list the outcomes: $S = \{s_1, s_2, \ldots\}$.
- If $A \subset S$ is an event, then A is also countable and (third axiom):
$$P(A) = P(\bigcup_{s_j \in A} \{s_j\}) = \sum_{s_j \in A} P(s_j)$$
- We sum the probabilities of the outcomes in the event to find the probability of the event.

Example

For $k \in \mathbb{N}$, I get $k - 2$ euro with probability $1/2^k$. Is this a (discrete) probability model?

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Finite Sample Space with Equally Likely Outcomes

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- A special case of a discrete probability model is a finite sample space where each outcome is equally likely: $S = \{s_1, s_2, \ldots, s_N\}$ where $P(s_i) = P(s_j)$, for all $i, j \in \{1, 2, \ldots, N\}$.
- Since all outcomes are equally likely, we have $P(s_i) = \frac{1}{N}$ for all $i \in \{1, 2, ..., N\}$.
- If A is an event with cardinality $|A| = M$, we have

$$P(A) = \sum_{s_j \in A} P(s_j) = \sum_{s_j \in A} \frac{1}{N} = \frac{M}{N} = \frac{|A|}{|S|}.$$

- So finding $P(A)$ reduces to a counting problem.[5]

[5] Counting methods are covered in Chapter 2 (which we will skip).

Exercises

Section 1.5: Exercises 13, 14, 15, 16, 17

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Conditional Probability

- How should we update probabilities of events, given more information?
- For example, say we know the probability of the event R, rain in Amsterdam on a random day, is 23%. Thus $P(R) = 0.23$.
- Suppose I pick a random day, but also say that it is cloudy on that given day. Now what is the probability that it rains, *given that* it is cloudy?
- If we denote the event cloudy by C, we want the **conditional probability** $P(R|C)$ i.e. the probability of R, *given that C has occurred*.

Example (to motivate the formal definition)

We roll a die. Let $A =$ "odd" $= \{1, 3, 5\}$ and $B =$ " ≤ 3" $= \{1, 2, 3\}$. Then $P(A) = |A|/|S| = 3/6 = 1/2$ and $P(A|B) = 2/3$ (intuitively). Observe that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

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Conditional Probability

Conditional probability. For $A, B \subset S$ with $P(B) > 0$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- The intuition behind the formula is that once we know B has occurred, the sample space reduces to B.
- $P(A|B)$ is undefined if $P(B) = 0$.
- $A \mapsto P(A|B)$ is a probability measure and has similar properties as $A \mapsto P(A)$ (e.g. $P(A^c|B) = 1 - P(A|B)$).

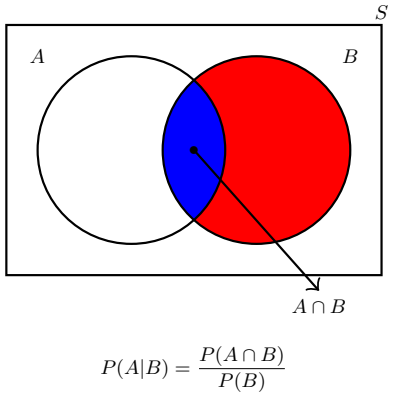

Figure: Venn diagram for the conditional probability $P(A|B)$.

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The Chain Rule for Conditional Probabilities

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$
$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1, A_2)$$

- Is useful if you know the conditional probabilities and want to know the probability of the intersection.
- Can be generalized to more than three events.

Example

In a factory there are 100 units of a product, 5 of which are defective. We pick 3 units at random. What is the probability that none of them is defect?

Solution: Let A_i = *"product i is not defect"* for $i = 1, 2, 3$. Then $P(A_1) = 95/100$, $P(A_2|A_1) = 94/99$, and $P(A_3|A_1, A_2) = 93/98$, hence

$$P(A_1 \cap A_2 \cap A_3) = (95/100) \times (94/99) \times (93/98).$$

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Independence

Let A be the event that it will rain tomorrow and let B be "heads" in a coin flip experiment. Suppose that $P(A) = 0.3$. What is $P(A|B)$? We believe that $P(A|B) = P(A) = 0.3$.

A and B are **independent** if $P(A \cap B) = P(A)P(B)$.

- If $P(B) > 0$, then A and B are independent if and only if $P(A|B) = P(A)$.
- Intuitively: The occurrence of B does not change the probability that A will occur (and *vice versa*).
- Independence can be generalized to more than 2 events (see textbook, page 47).
- In practice, independence is often clear from the context or can be assumed.

Example.

I toss a fair coin until I observe the first tails at which I stop. Let X be the total number of tosses. Find $P(X = 5)$.

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Exercise (Example 1.20 in textbook)

I pick a random number from $\{1, 2, \ldots, 10\}$ and call it N. Suppose that all outcomes are equally likely. Let A be the event that N is less than 7, and let B be the event that N is an even number. Are A and B independent? And what about if A is the event that N is less than 6? Can you explain the results?

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Differences between Disjointedness and Independence

Concept	Meaning	Formulas
Disjoint	A and B cannot occur at the same time	$A \cap B = \emptyset$
Independent	A gives no information about B	$P(A\|B) = P(A)$, $P(B\|A) = P(B)$

If $P(A), P(B) > 0$, then disjointness implies dependence:

Suppose that $A \cap B = \emptyset$, then $P(A \cap B) = P(\emptyset) = 0$, but $P(A)P(B) > 0$, and therefore $P(A \cap B) \neq P(A)P(B)$, so A and B are dependent.

Intuition: Knowing that A occurred, tells us something about B, namely that B will *not* occur.

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Useful Results for Independent Events

- If A and B are independent then
  - A and B^C are independent.
  - A^C and B are independent.
  - A^C and B^C are independent.
- If $A_1, A_2, \ldots, A_n$ are independent, then $P(A_1 \cup A_2 \cup \ldots \cup A_n) = 1 - (1 - P(A_1))(1 - P(A_2)) \ldots (1 - P(A_n))$.

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The Law of Total Probability

For any partition $B_1, B_2, \ldots$ of the sample space S and event A:

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i)$$

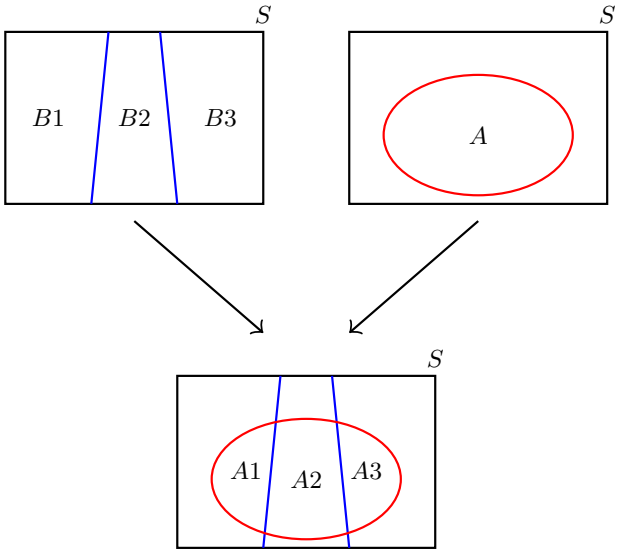

Figure: Law of total probability

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The Law of Total Probability

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Proof: $S = \cup_i B_i$ and therefore

$$A = A \cap S = A \cap (\cup_i B_i) = \cup_i (A \cap B_i)$$

but the B_i's are disjoint thus so are the $A \cap B_i$'s and therefore

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i).$$

$\square$

The Law of Total Probability

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Example

I have 3 bags of 100 marbles. Bag 1 contains 75 red ones and 25 blue ones, bag 2 contains 60 red ones and 40 blue ones, and bag 3 contains 45 red ones and 55 blue ones. I choose a random bag and take a random marble from it. What is the probability that the chosen marble is red?

Solution: Let $R =$ "chosen marble is red" and $B_i =$ "choose bag i" for $i = 1, 2, 3$. We know that $P(B_i) = 1/3$, $P(R|B_1) = 0.75$, $P(R|B_2) = 0.60$, $P(R|B_1) = 0.45$, and therefore

$$P(A) = \sum_{i=1}^{3} P(A|B_i)P(B_i) = (0.75 + 0.60 + 0.45)/3 = 0.60.$$

Bayes' Rule

Suppose we know $P(A|B)$ but want to know $P(B|A)$:

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}.$$

Bayes' rule. For all events A and B with $P(A) > 0$:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

For any partition $B_1, B_2, \ldots$ of the sample space S and event A with $P(A) > 0$:

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}.$$ [a]

[a] Here we used the law of total probability: $P(A) = \sum_i P(A|B_i)P(B_i)$.

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Bayes' rule (example)

The false positive paradox

A certain disease affects about 1 out of 10.000 people. There is a test for this disease, which is quite accurate: The probability of a false positive is 2% and the probability of a false negative is 1%. A random person is tested and the result is positive. What is the probability that the person has the disease?

Solution: Let $T =$ "tests positive" and $D =$ "has the disease". We know that $P(D) = 0.0001$, $P(T|D^c) = 0.02$, and $P(T^c|D) = 0.01$. Want to know $P(D|T)$:

$$\begin{aligned} P(D|T) &= \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)} \\ &= \frac{(1-0.01)\times 0.0001}{(1-0.01)\times 0.0001 + 0.02\times(1-0.0001)} \\ &= 0.0049 \text{ (less than 0.5\%!)} \end{aligned}$$

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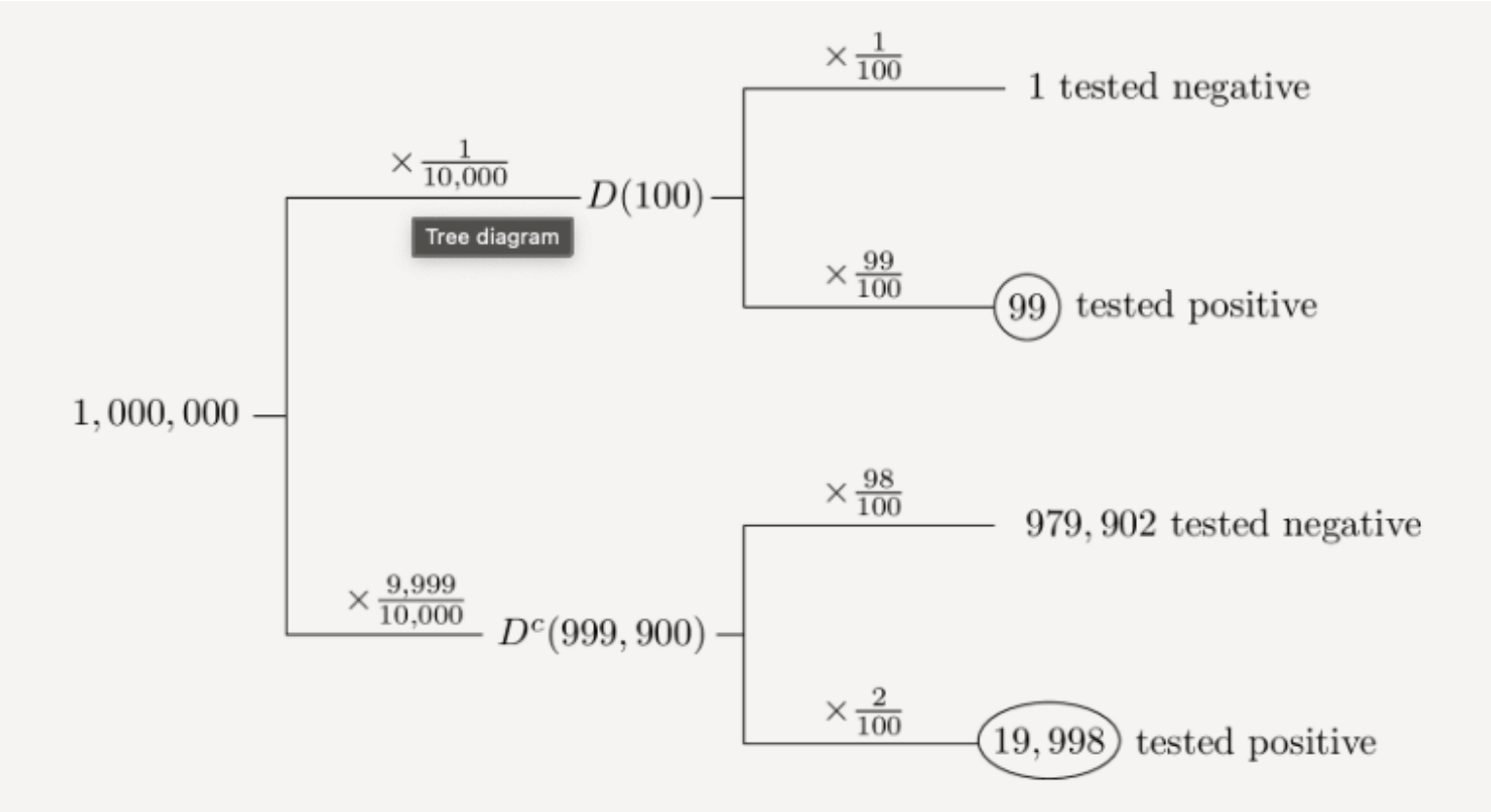

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